

Research Interests

I work on optimizing and chaining LLM-based query operators within dataflow systems, with a focus on scalable support for high-dimensional vector processing, contextual embeddings, and RAG frameworks.

Education

- **Brown University**, Providence, RI, United States
Doctor of Philosophy, Computer Science 2023/09 – Expected 2028
– Committee: Uğur Çetintemel (advisor), Malte Schwarzkopf, Andrew Crotty.
- **Simon Fraser University**, Burnaby, BC, Canada
Bachelor of Science, Computer Science 2019/01 – 2022/05
– Advisor: Tianzheng Wang

Publications

Google Scholar: Duo Lu

- [An In-Depth Study of Filter-Agnostic Vector Search on a PostgreSQL Database System.](#)
Duo Lu, H. Caminal, M. Chatzakis, Y. Papakonstantinou, Y. Chronis, V. Jain, F. Ozcan
International Conference on Management of Data (SIGMOD), 2026. (To appear)
- [Making Prompts First-Class Citizens for Adaptive LLM Pipelines](#)
Uğur Çetintemel, Shu Chen, Alexander W. Lee, Deepti Raghavan, Duo Lu, Andrew Crotty
The Conference on Innovative Data Systems Research (CIDR), 2026.
- [VectraFlow: Integrating Vectors into Stream Processing.](#)
Duo Lu, Siming Feng, Jonathan Zhou, Franco Solleza, Malte Schwarzkopf, Uğur Çetintemel
The Conference on Innovative Data Systems Research (CIDR), 2025.
- [Evaluating Persistent Memory Range Indexes: Part Two.](#)
Yuliang He, Duo Lu, Kaisong Huang, Tianzheng Wang
International Conference on Very Large Data Bases (Proceedings of VLDB), 2022.

Experience

- **Google** Sunnyvale, CA
Student Researcher, SystemsResearch@Google (SRG) 2025/05 – 2025/12
Advisors: Helena Caminal, Yannis Chronis, Fatma Ozcan
Design and evaluate filtered vector search algorithms; closely collaborate with Google AlloyDB team.
- **Brown University** Providence, RI
Research Assistant, Brown Database Group 2023/09 – Present
Advisor: Uğur Çetintemel
Teaching Assistant, Department of Computer Science
 - CSCI 2270: Data Management for AI Spring 2025
 - CSCI 1270: Database Management Systems Fall 2024
- **Fortinet** Vancouver, BC
Embedded Software Developer (FortiOS, router team) 2022/05 – 2023/08
Maintained and resolved bugs for the open-sourced operating system (FortiOS), focusing on enterprise dynamic routing protocols such as BGP, OSPF, RIP, and ISIS.

References

- **Dr. Uğur Çetintemel** – Professor, Brown University
- **Dr. Tianzheng Wang** – Associate Professor, Simon Fraser University
- **Dr. Martin Ester** – Professor, Simon Fraser University

ugur_cetintemel@brown.edu
tzwang@sfu.ca
ester@sfu.ca